UNIT 2 · TOPIC 3(A)

Farm Machine System Characteristics

Syllabus topic: Farm machine system characteristics and evaluation

The farm machine as a system

A farm machine is a system of functional components arranged so that material (soil, seed, crop) flows through a sequence of unit operations. Kepner (Ch. 1) treats design as meeting functional requirements: each component performs either a **processing** function (cutting, metering, threshing, separating, cleaning) or a **transport/handling** function (gathering, conveying, elevating, discharging).

Example — a grain combine as a system:

1. Gathering
2. Cutting
3. Feeding
4. Threshing
5. Separating
6. Cleaning
7. Handling

If any one component fails or is mismatched (e.g., feeding), it limits the whole system: **system capacity is set by the weakest (limiting) component**.

Stages of farm machinery development

1. Ideation and definition of functional requirements
2. Design and analysis
3. Prototype fabrication
4. Laboratory testing
5. Field testing (multiple seasons/locations)
6. Refinement
7. Commercial production
8. Feedback

Kepner Ch. 1 — research and development sequence; also the AICRP-FIM prototype development route in India.

Machine system characteristics

Load character	Example
Steady	Pump drive
Fluctuating	Plough draft varies ±25–50% about the mean
Cyclic	Baler plunger
Shock	Rotavator striking stones

Size service factors and overload devices to the load character.

- **Draft–speed characteristic of tillage:** D = a + bV² — draft rises with the square of speed for mouldboard bodies. Match this implement demand curve to the power source's capability (full treatment in Topic 5 and Unit 3).
- **Torque–speed demand of rotary machines** (chaff cutter, thresher): high starting torque + flywheel effect; the flywheel smooths power peaks.
- **Timeliness:** the value of an operation depends on finishing it within the optimum period; timeliness cost enters machine selection (Unit 3).
- **Reliability:** the probability that a machine performs its function for a stated period under stated conditions.

$$R_{system} = R_1 \times R_2 \times \ldots \times R_n \text{ (series system — declines rapidly as components increase)}$$

Practice MCQs

Q41. The capacity of a multi-component machine system (e.g., combine) is determined by:

(a) The largest component
(b) The limiting (weakest) functional component
(c) The engine alone
(d) Average of all components

Q42. If a machine has 4 components in series, each with reliability 0.95, system reliability is about:

(a) 0.95
(b) 0.99
(c) 0.81
(d) 0.86

Q43. Draft of a mouldboard plough varies with forward speed as:

(a) D = a + bV
(b) D = a + bV²
(c) D = a/V
(d) D = constant

Q74. Timeliness of operation in machinery system evaluation refers to:

(a) Machine speed
(b) Completing the operation within the optimum period to avoid yield/quality loss
(c) Idle time
(d) Shift hours

Answers

Q41 — (b) The limiting (weakest) functional component

Q42 — (c) 0.81 — *$0.95^4 = 0.8145$ — series reliability multiplies; why simple, robust designs matter in farm machines.*

Q43 — (b) D = a + bV²

Q74 — (b) Completing the operation within the optimum period to avoid yield/quality loss

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 1 (Research & Development), Ch. 2 (Field Capacities), §14.11 (Reciprocating Unbalance in a Mower); theory-of-machines standards (balancing, flywheels); tractor–implement mechanics references.